

Supplementary Appendix

This appendix has been provided by the authors to give readers additional information about their work.

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Supplementary Materials for

Apalutamide in Metastatic Castration-Sensitive Prostate Cancer

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SUPPLEMENTARY METHODS

Definitions of Secondary End Points

- Time to cytotoxic chemotherapy was defined as the time from randomization to initiation of cytotoxic chemotherapy.
- Time to pain progression was defined as the time from randomization to pain progression (average increase in 2 points from baseline in Brief Pain Inventory-Short Form [BPI-SF] worst pain intensity observed at two consecutive evaluations ≥ 3 weeks apart, with an average worst pain score of >4 in patients who have had no decrease in opioids or initiation of chronic opioids, whichever occurs first).
 - BPI-SF worst pain (item 3) was used for the time to pain progression end point.
 - Scores range from 0 to 10, with lower scores representing lower levels of pain intensity; a change of 2 was the minimally important difference.¹
- Time to chronic opioid use was defined as the time from randomization to chronic opioid use.
 - Chronic opioid use was defined as administration of opioid analgesics for ≥ 3 weeks for oral or ≥ 7 days for nonoral formulations.
 - For patients who were already receiving opioids at study entry, chronic opioid use was defined as a $\geq 30\%$ increase in total daily dose of the opioid analgesics lasting for ≥ 3 weeks for oral or ≥ 7 days for nonoral formulations.
 - Administration of as-needed (eg, not fixed or scheduled dosage) opioid analgesics or extended opioid use for treatment other than the patient's prostate cancer did not require discontinuation from study treatment.
- Time to skeletal-related event was defined as the time from randomization to first observation of a skeletal-related event (symptomatic pathologic fracture, spinal cord compression, radiation to bone, or surgery to bone).

Definitions of Exploratory End Points

- Time to PSA progression was defined as the time from randomization to date of PSA progression based on Prostate Cancer Working Group 2 criteria.²
- Time to second progression-free survival was defined as the time from randomization to first occurrence of investigator-determined disease progression (PSA progression, progression on imaging, or clinical progression) while patient was receiving first subsequent therapy for prostate cancer, or death due to any cause, whichever occurs first.
- Time to symptomatic local progression was defined as the time from randomization to date of symptomatic local progression.

Prostate Cancer Working Group 2 Criteria

Patients in TITAN were assessed for efficacy per modified Response Evaluation Criteria In Solid Tumors version 1.1 and Prostate Cancer Working Group 2 criteria.² Prostate Cancer Working Group 2 criteria for progression are as follows:

- PSA: first $\geq 25\%$ increase from baseline and ≥ 2 ng/mL above the nadir (confirmed by second value 3 or more weeks later)
- Soft-tissue lesions: follow Response Evaluation Criteria in Solid Tumors with caveats
 - Imaging should include a computed tomography scan or magnetic resonance imaging at a minimum; centers with relevant expertise should utilize endorectal magnetic resonance imaging or transrectal ultrasound
 - Only report changes in lymph nodes that were ≥ 2 cm in diameter at baseline
 - Record changes in nodal and visceral soft tissue sites separately
 - Record complete elimination of disease at any site separately
 - Confirm favorable change with a second scan
 - Record changes using a waterfall plot

- Progression at first assessment must be confirmed by a second scan ≥ 6 weeks later
- Bone: appearance of ≥ 2 new lesions, and, for the first reassessment only, a confirmatory scan performed ≥ 6 weeks later indicating a minimum of ≥ 2 new lesions

SUPPLEMENTARY RESULTS

Rash Management

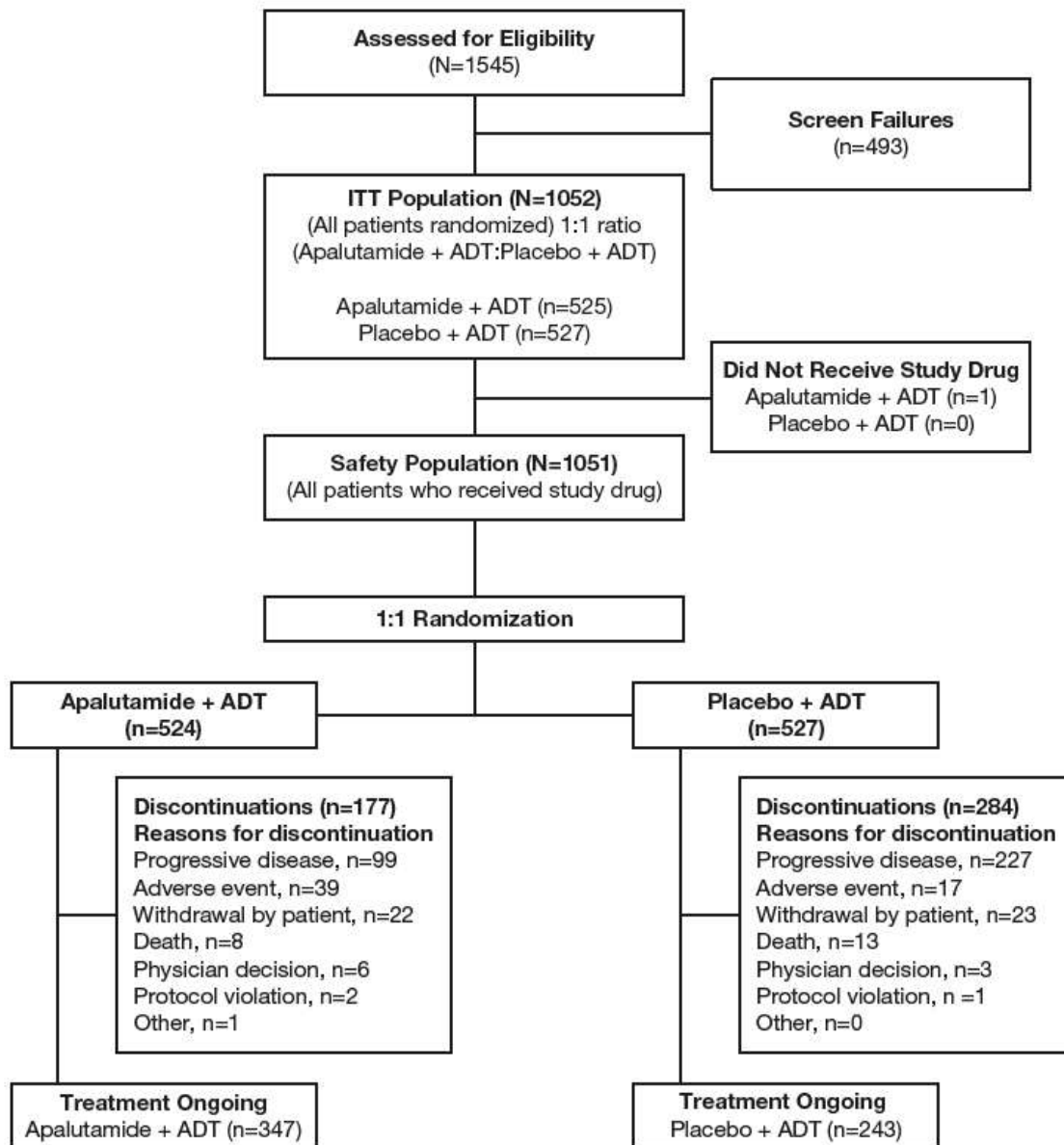
Skin rash associated with apalutamide was commonly described as generalized or maculo-papular.

Treatment-emergent skin rash was reported by 27.1% of patients in the apalutamide group versus 8.5% in the placebo group. Grade 3 rashes were reported with apalutamide (6.3%) and placebo (0.6%) treatment.

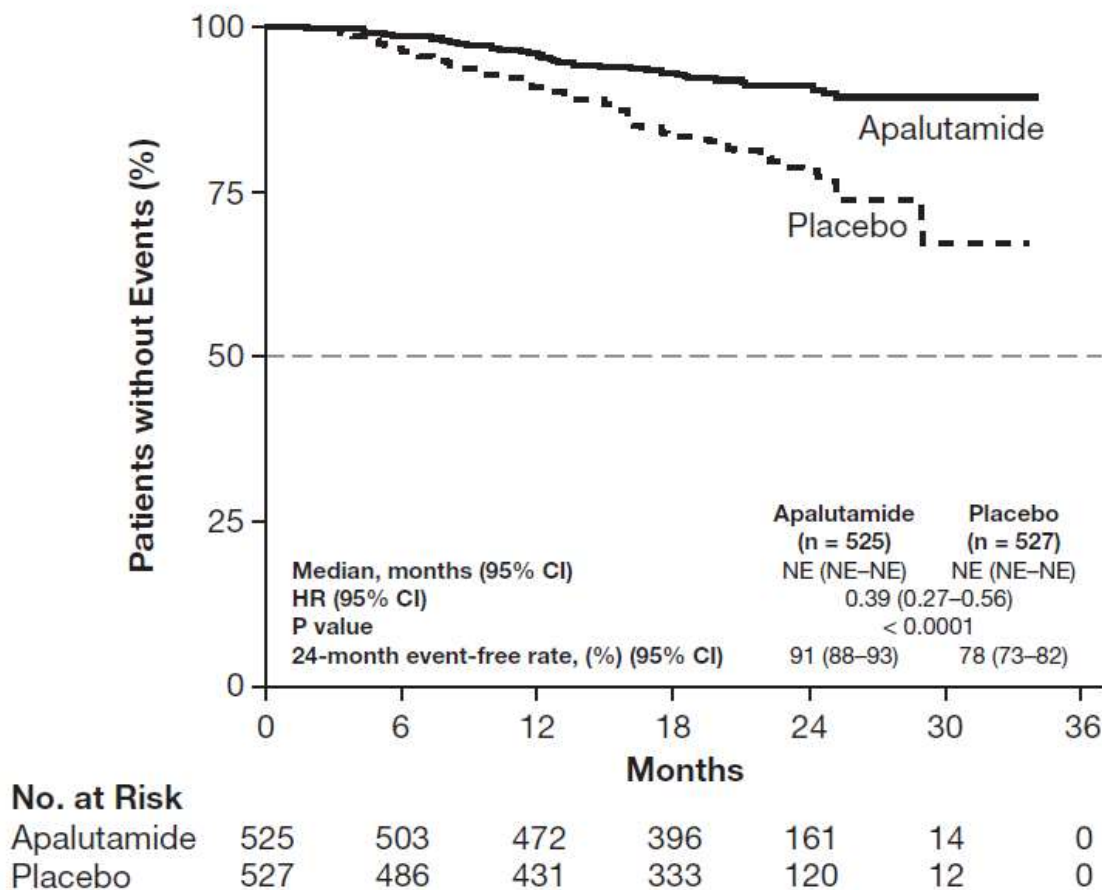
No Stevens-Johnson syndrome or toxic epidermal necrolysis was reported. Skin rash led to treatment discontinuation, dose reduction, and dose interruption in 12 (2.3%), 28 (5.3%), and 44 (8.4%), patients, respectively, in the apalutamide group, and 1 (0.2%), 4 (0.8%), and 5 (0.9%), patients, respectively, in the placebo group. Grade ≥ 3 rash led to discontinuation in 7 (1.3%) and 1 (0.2%) patients in the apalutamide and placebo groups, respectively (Table S4, Supplementary Appendix). For patients who had a skin rash, treatment included topical corticosteroids, oral antihistamines, systemic corticosteroids, drug interruption, and dose reduction. Median time to onset of rash was 81 days in the apalutamide group and 141 days in the placebo group.

SUPPLEMENTARY FIGURES

Supplementary Figure S1. CONSORT Diagram.

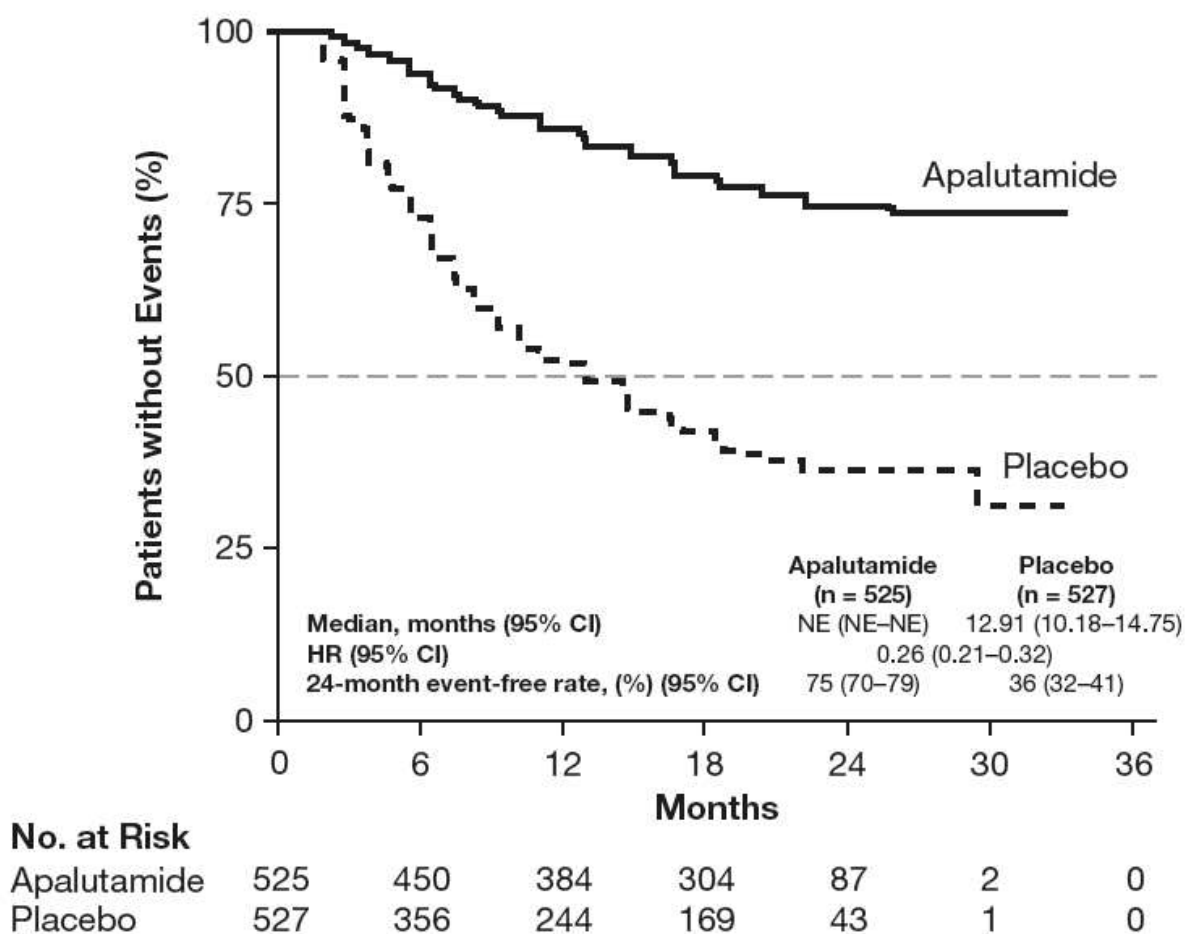


Time to Initiation of Cytotoxic Chemotherapy



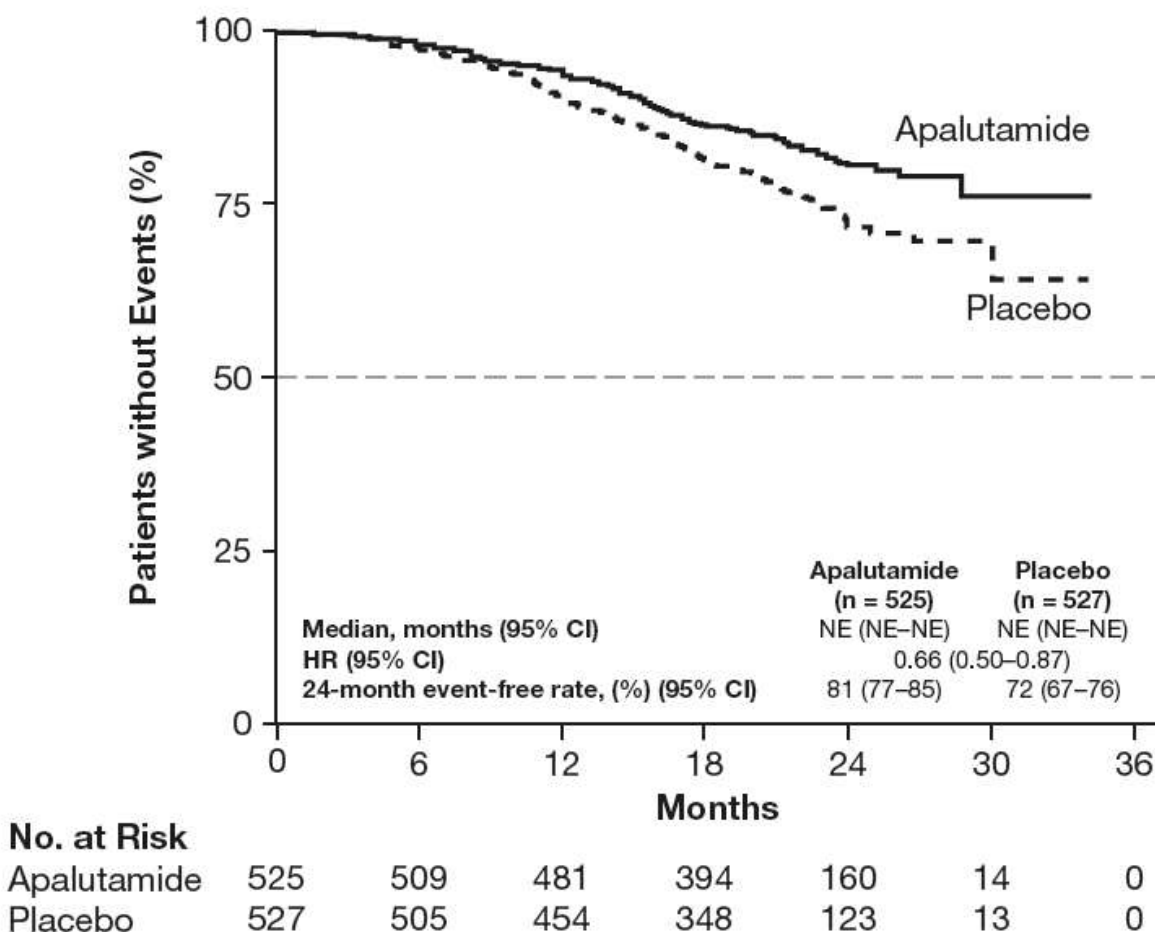
Supplementary Figure S2. **Kaplan-Meier Estimate of Time to Cytotoxic Chemotherapy.** Analyses were performed using a log-rank test with stratification by Gleason score at diagnosis (≤ 7 vs. > 7), region (North America and European Union vs. all other countries), and prior treatment with docetaxel (yes vs. no).

Time to PSA Progression



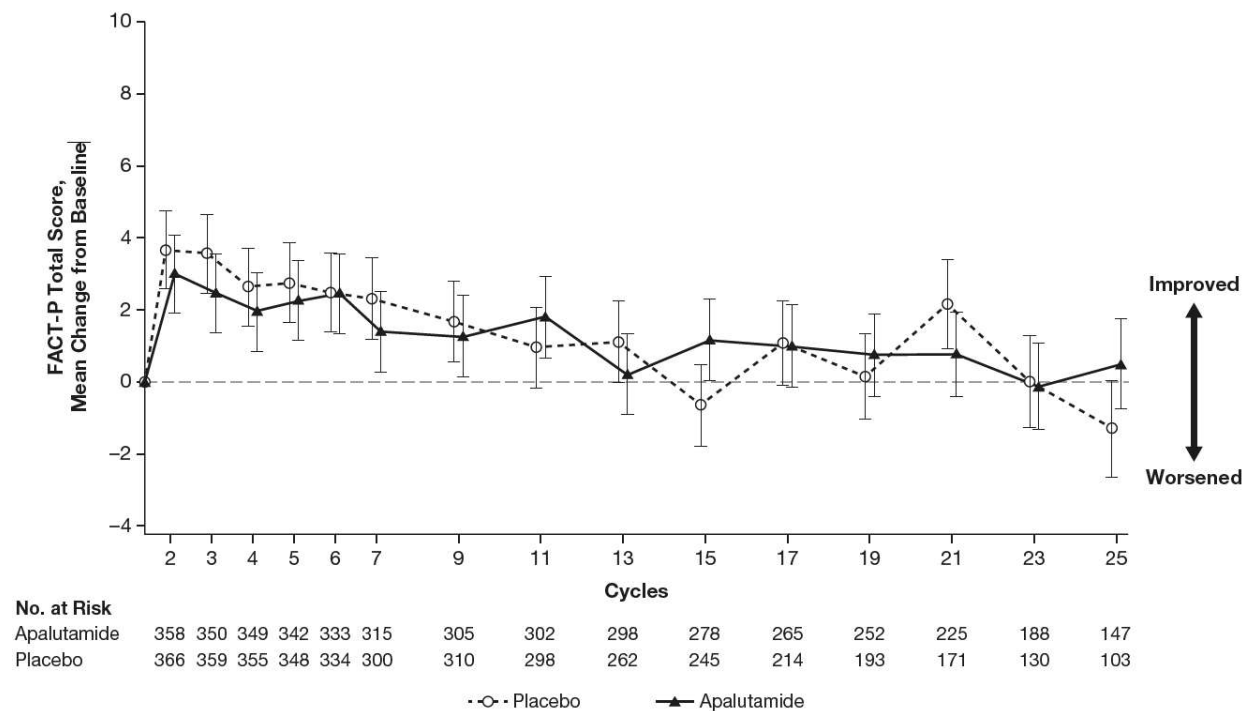
Supplementary Figure S3. **Kaplan-Meier Estimate of Time to PSA Progression.** Time to PSA progression was the time from date of randomization to date of PSA progression, based on Prostate Cancer Working Group 2 criteria. Analyses were performed using a log-rank test with stratification by Gleason score at diagnosis (≤ 7 vs. > 7), region (North America and European Union vs. all other countries), and prior treatment with docetaxel (yes vs. no).

Time to Second Progression-Free Survival



Supplementary Figure S4. **Kaplan-Meier Estimate of Time to Second Progression-free Survival.**

Second progression-free survival was the time from date of randomization to first occurrence of investigator-determined disease progression (PSA progression, progression on imaging, or clinical progression) while patient was receiving first subsequent therapy for prostate cancer or death due to any cause, whichever occurs first. Analyses were performed using a log-rank test with stratification by Gleason score at diagnosis (≤ 7 vs. > 7), region (North America and European Union vs. all other countries), and prior treatment with docetaxel (yes vs. no).



Supplementary Figure S5. **Repeated-Measures Mixed Effects Analyses for Mean Change from Baseline in FACT-P Total Score.** Error bars are standard errors of the mean. Raw FACT-P scores range from 0 to 156, with higher scores indicating more favorable health-related quality of life; a 6- to 10-point change in FACT-P total score would be the minimally important difference. However, this figure presents mean changes in total scores compared with baseline rather than raw total scores. FACT-P, Functional Assessment of Cancer Therapy-Prostate.

Supplementary Table S1. **Demographics and Baseline Disease Characteristics, Expanded List.**

	Apalutamide (n=525)	Placebo (n=527)
Age, median – yr (range)	69 (45–94)	68 (43–90)
Age (yr) – no. (%)		
<65	149 (28.4)	182 (34.5)
65–69	136 (25.9)	108 (20.5)
70–74	107 (20.4)	124 (23.5)
≥75	133 (25.3)	113 (21.4)
Race – no. (%)		
White	354 (67.4)	365 (69.3)
Asian	119 (22.7)	110 (20.9)
Black or African American	10 (1.9)	9 (1.7)
American Indian or Alaska native	6 (1.1)	13 (2.5)
Not reported	11 (2.1)	8 (1.5)
Other	24 (4.6)	22 (4.2)
Multiple	1 (0.2)	0
ECOG PS score – no. (%)	n=525	n=527
0	328 (62.5)	348 (66.0)
1	197 (37.5)	178 (33.8)
2	0	1 (0.2)
Gleason score at initial diagnosis – no. (%)	n=525	n=527
<7	41 (7.8)	39 (7.4)
7	133 (25.3)	130 (24.7)
>7	351 (66.9)	358 (67.9)
Tumor, lymph node, metastatic stage at initial diagnosis – no. (%)	n=525	n=527
T0	1 (0.2)	0
T1	41 (7.8)	27 (5.1)
T2	146 (27.8)	110 (20.9)
T3	210 (40.0)	225 (42.7)
T4	76 (14.5)	105 (19.9)
TX	51 (9.7)	60 (11.4)
N0	212 (40.4)	216 (41.0)
N1	199 (37.9)	184 (34.9)
NX	114 (21.7)	127 (24.1)
M0	85 (16.2)	59 (11.2)
M1	411 (78.3)	441 (83.7)
MX	29 (5.5)	27 (5.1)
Disease volume – no. (%)		
Low	200 (38.1)	192 (36.4)
High	325 (61.9)	335 (63.6)
Extent of disease at study entry – no. (%)		
Bone	525 (100.0)	527 (100.0)
Bone only	289 (55.0)	269 (51.0)
Lymph node	199 (37.9)	219 (41.6)
Visceral and bone	56 (10.7)	72 (13.7)
Lung	47 (9.0)	64 (12.1)
Liver	12 (2.3)	13 (2.5)

Soft tissue and bone	22 (4.2)	27 (5.1)
Median time from initial diagnosis to randomization (range) – mo	4.1 (0.5–222.9)	4.0 (0.7–341.4)
Prior docetaxel – no. (%)		
Yes*	58 (11.0)	55 (10.4)
No	467 (89.0)	472 (89.6)
Median no. of cycles of docetaxel among patients with prior docetaxel	6	6
Prior therapy for localized prostate cancer – no. (%)		
Prostatectomy or radiotherapy	94 (17.9)	79 (15.0)
Prostatectomy only	26 (5.0)	27 (5.1)
Radiotherapy only	47 (9.0)	39 (7.4)
Both prostatectomy and radiotherapy	21 (4.0)	13 (2.5)
Median PSA (range) – µg/L	5.97 (0–2682)	4.02 (0–2229)
Median LDH (range) – U/L	178 (88–1248)	179 (85–1514)
Median alkaline phosphatase (range) – U/L	98 (26–5193)	94 (32–3892)
Mean baseline BPI-SF pain score† – no. (%)	198 (37.7)	200 (38.0)
0 (no pain)	195 (37.1)	207 (39.3)
1 to 3 (mild pain)	98 (18.7)	95 (18.0)
4 to 7 (moderate pain)	12 (2.3)	11 (2.1)
8 to 10 (severe pain)		
Median baseline FACT-P total score‡	113	113

ECOG PS, Eastern Cooperative Oncology Group performance status; FACT-P, Functional Assessment of Cancer Therapy-Prostate; PSA, prostate-specific antigen.

* 27 patients (46.6%) in the apalutamide group and 22 patients (40.0%) in the placebo group were N1 at diagnosis.

† Scores range 0 to 10, with lower scores representing lower levels of pain intensity; a change of 2 was the minimally important difference.¹

‡ Scores range from 0 to 156, with higher scores indicating more favorable health-related quality of life. A 6- to 10-point change in FACT-P total score is the minimal important difference.

Supplementary Table S2. **Prior Prostate Cancer Therapy.**

	Apalutamide	Placebo
	(n=525)	(n=527)
Prostatectomy or radiotherapy – no. (%)	94 (17.9)	79 (15.0)
Prostatectomy only	26 (5.0)	27 (5.1)
Radiotherapy only	47 (9.0)	39 (7.4)
Both prostatectomy and radiotherapy	21 (4.0)	13 (2.5)
Hormonal therapy – no. (%)	525 (100)	527 (100)
First-generation antiandrogen	352 (67.0)	361 (68.5)
Gonadotropin-releasing hormone agonist	462 (88.2)	455 (86.3)
Gonadotropin-releasing hormone antagonist	56 (10.7)	53 (10.1)
Bilateral orchiectomy	33 (6.3)	40 (7.6)
Docetaxel – no. (%)	58 (11.0)	55 (10.4)
Vandetanib – no. (%)	1 (0.2)	0

Supplementary Table S3. **First Subsequent Systemic Prostate Cancer Therapy.**

	Apalutamide	Placebo
	(n=525)	(n=527)
Patients who discontinued treatment for any reason* and remained alive (<i>denominator for first subsequent therapy calculations below</i>) – no.	170	271
Patients who discontinued because of disease progression – no. (%)	99 (18.9)	227 (43.1)
Patients with adverse events leading to discontinuation – no. (%)	42 (8.0)	28 (5.3)
Patients with life-prolonging subsequent therapy for prostate cancer† – no. (%)	64 (37.6)	165 (60.9)
Patients receiving subsequent systemic prostate cancer therapy at data cutoff‡ – no. (%)	87 (51.2)	190 (70.1)
First subsequent therapy		
Hormonal therapy – no. (%)	44 (25.9)	98 (36.2)
Abiraterone acetate plus prednisone	21 (12.4)	45 (16.6)
Bicalutamide	16 (9.4)	31 (11.4)
Enzalutamide	3 (1.8)	17 (6.3)
Other§	4 (2.4)	5 (1.8)
Chemotherapy – no. (%)	35 (20.6)	73 (26.9)
Docetaxel	29 (17.1)	67 (24.7)
Cabazitaxel	1 (0.6)	2 (0.7)
Other**	5 (2.9)	4 (1.5)

Other therapy – no. (%)	8 (4.7)	19 (7.0)
Radium-223	2 (1.2)	4 (1.5)
Sipuleucel-T	2 (1.2)	5 (1.8)
Other***	4 (2.4)	10 (3.7)

* Possible reasons for discontinuation were disease progression, adverse event, withdrawal by patient, death, physician decision, and protocol violation.

† Included docetaxel, abiraterone acetate plus prednisone, enzalutamide, cabazitaxel, radium-223, and sipuleucel-T.

‡ Some patients were unblinded after discontinuation and prior to first subsequent therapy to allow enrollment in a subsequent clinical trial.

§ Included diethylstilbestrol, flutamide, and cyproterone.

** Included etoposide, paclitaxel, estramustine, carboplatin, and cisplatin.

*** Included zoledronic acid, clodronate, prednisolone, and prednisone.

Supplementary Table S4. **Treatment-Emergent Adverse Events Leading to Treatment**

Discontinuation, Dose Reduction, and Dose Interruption.

	Apalutamide		Placebo	
	(n=524)		(n=527)	
	All Grades	Grade ≥3	All Grades	Grade ≥3
Discontinuation				
Patients with treatment-emergent adverse events leading to discontinuation – no. (%)	42 (8.0)	26 (5.0)	28 (5.3)	22 (4.2)
Rash*	12 (2.3)	7 (1.3)	1 (0.2)	1 (0.2)
New cancer neoplasm†	7 (1.3)	5 (1.0)	5 (0.9)	3 (0.6)
Infections‡	1 (0.2)	1 (0.2)	3 (0.6)	0
Fatigue	4 (0.8)	1 (0.2)	0	0
Ischemic cardiac events§	2 (0.4)	2 (0.4)	2 (0.4)	2 (0.4)
Seizure	2 (0.4)	1 (0.2)	1 (0.2)	0
Pulmonary embolism	1 (0.2)	1 (0.2)	2 (0.4)	1 (0.2)
Sudden death	0	0	2 (0.4)	2 (0.4)
Cerebrovascular disorder	1 (0.2)	1 (0.2)	0	0
Parkinson's disease	1 (0.2)	1 (0.2)	0	0
Radiculopathy	1 (0.2)	1 (0.2)	0	0
Cognitive disorder	0	0	1 (0.2)	0

Intracranial hemorrhage	0	0	1 (0.2)	1 (0.2)
Loss of consciousness	0	0	1 (0.2)	1 (0.2)
Atrial fibrillation	1 (0.2)	1 (0.2)	0	0
Cardio-respiratory arrest	1 (0.2)	1 (0.2)	0	0
Cardiogenic shock	1 (0.2)	1 (0.2)	0	0
Cardiac failure	0	0	1 (0.2)	1 (0.2)
Sudden cardiac death	1 (0.2)	1 (0.2)	0	0
Hypothermia	0	0	1 (0.2)	1 (0.2)
Anxiety	1 (0.2)	0	0	0
Euphoric mood	1 (0.2)	0	0	0
Suicide	0	0	1 (0.2)	1 (0.2)
Large intestinal ulcer perforation	1 (0.2)	1 (0.2)	0	0
Lip swelling	1 (0.2)	0	0	0
Dyspnea	0	0	1 (0.2)	0
Pleural effusion	0	0	1 (0.2)	1 (0.2)
Pulmonary edema	0	0	1 (0.2)	1 (0.2)
Respiratory failure	0	0	1 (0.2)	1 (0.2)
Dehydration	1 (0.2)	0	0	0

Hot flush	1 (0.2)	0	0	0
Subdural hemorrhage	0	0	1 (0.2)	1 (0.2)
Alanine aminotransferase increased	0	0	1 (0.2)	1 (0.2)
	Dose Reduction			
Patients with treatment-emergent adverse events leading to dose reduction – no. (%)	37 (7.1)	19 (3.6)	11 (2.1)	1 (0.2)
Rash*	28 (5.3)	11 (2.1)	4 (0.8)	1 (0.2)
Aspartate aminotransferase or alanine aminotransferase increased	0	0	4 (0.8)	0
Fatigue	2 (0.4)	2 (0.4)	0	0
Headache	1 (0.2)	0	1 (0.2)	0
Hypertension	1 (0.2)	1 (0.2)	1 (0.2)	0
Neutropenia	2 (0.4)	2 (0.4)	0	0
Cognitive disorder	1 (0.2)	1 (0.2)	0	0
General physical health deterioration	1 (0.2)	1 (0.2)	0	0
Hot flush	1 (0.2)	0	0	0
Asthenia	0	0	1 (0.2)	0
Dehydration	1 (0.2)	0	0	0
Lethargy	0	0	1 (0.2)	0

Pain**	1 (0.2)	0	0	0
Performance status decreased	0	0	1 (0.2)	0
Vertigo	1 (0.2)	1 (0.2)	0	0
Weight decreased	1 (0.2)	1 (0.2)	0	0
	Dose Interruption			
Patients with treatment-emergent events leading to dose interruption – no. (%)	104 (19.8)	68 (13.0)	63 (12.0)	43 (8.2)
Rash*	44 (8.4)	20 (3.8)	5 (0.9)	2 (0.4)
Aspartate aminotransferase or alanine aminotransferase increased	7 (1.3)	4 (0.8)	12 (2.3)	7 (1.3)
Pain**	9 (1.7)	4 (0.8)	7 (1.3)	3 (0.6)
Infections‡	6 (1.1)	4 (0.8)	9 (1.7)	3 (0.6)
Hypertension	6 (1.1)	6 (1.1)	6 (1.1)	6 (1.1)
Fatigue	6 (1.1)	1 (0.2)	2 (0.4)	1 (0.2)
Vomiting	3 (0.6)	0	4 (0.8)	0
Anemia	2 (0.4)	0	3 (0.6)	2 (0.4)
Asthenia	3 (0.6)	3 (0.6)	2 (0.4)	1 (0.2)
Decreased appetite	2 (0.4)	1 (0.2)	3 (0.6)	1 (0.2)
New cancer neoplasm†	3 (0.6)	1 (0.2)	1 (0.2)	1 (0.2)

Diarrhea	1 (0.2)	0	3 (0.6)	0
Chronic obstructive pulmonary disease	2 (0.4)	2 (0.4)	2 (0.4)	2 (0.4)
Pyrexia	3 (0.6)	0	1 (0.2)	0
Urinary retention	2 (0.4)	2 (0.4)	2 (0.4)	2 (0.4)
Arthralgia	2 (0.4)	0	1 (0.2)	0
Insomnia	3 (0.6)	0	0	0
Acute kidney injury	1 (0.2)	1 (0.2)	1 (0.2)	1 (0.2)
Anxiety	2 (0.4)	0	0	0
Cerebrovascular accident	1 (0.2)	1 (0.2)	1 (0.2)	0
Dizziness	1 (0.2)	0	1 (0.2)	0
Dyspnea	1 (0.2)	0	1 (0.2)	1 (0.2)
Fall	1 (0.2)	1 (0.2)	1 (0.2)	1 (0.2)
Headache	1 (0.2)	0	1 (0.2)	1 (0.2)
Hematuria	2 (0.4)	2 (0.4)	0	0
Hot flush	1 (0.2)	0	1 (0.2)	0
Hypersensitivity	1 (0.2)	0	1 (0.2)	0
Hypertriglyceridemia	2 (0.4)	2 (0.4)	0	0
Hyponatremia	1 (0.2)	1 (0.2)	1 (0.2)	1 (0.2)

Muscular weakness	1 (0.2)	0	1 (0.2)	1 (0.2)
Nausea	1 (0.2)	0	1 (0.2)	0
Neutropenia	2 (0.4)	2 (0.4)	0	0
Spinal cord compression	1 (0.2)	1 (0.2)	1 (0.2)	1 (0.2)
Acute respiratory failure	1 (0.2)	1 (0.2)	0	0
Allergy to chemicals	1 (0.2)	1 (0.2)	0	0
Angina unstable	0	0	1 (0.2)	1 (0.2)
Atrial fibrillation	1 (0.2)	1 (0.2)	0	0
Bile duct obstruction	0	0	1 (0.2)	1 (0.2)
Blood alkaline phosphatase increased	1 (0.2)	1 (0.2)	0	0
Cardiac amyloidosis	0	0	1 (0.2)	1 (0.2)
Cardiac disorder	0	0	1 (0.2)	1 (0.2)
Cardiac failure	0	0	1 (0.2)	0
Cardiac failure congestive	1 (0.2)	1 (0.2)	0	0
Cheilitis	1 (0.2)	0	0	0
Chronic hepatitis	1 (0.2)	0	0	0
Cognitive disorder	1 (0.2)	1 (0.2)	0	0
Confusional state	1 (0.2)	1 (0.2)	0	0

Constipation	1 (0.2)	0	0	0
Dehydration	0	0	1 (0.2)	0
Depression	1 (0.2)	0	0	0
Diabetes mellitus	1 (0.2)	1 (0.2)	0	0
Diplegia	0	0	1 (0.2)	1 (0.2)
Dysuria	1 (0.2)	1 (0.2)	0	0
Edema	1 (0.2)	0	0	0
Eyelid edema	1 (0.2)	0	0	0
Facial edema	1 (0.2)	0	0	0
Febrile neutropenia	1 (0.2)	1 (0.2)	0	0
Gamma-glutamyltransferase increased	1 (0.2)	1 (0.2)	0	0
Gastric ulcer perforation	1 (0.2)	1 (0.2)	0	0
Gastritis	0	0	1 (0.2)	1 (0.2)
General physical health deterioration	1 (0.2)	1 (0.2)	0	0
Hallucination	1 (0.2)	0	0	0
Hematoma	1 (0.2)	1 (0.2)	0	0
Hemorrhagic erosive gastritis	1 (0.2)	1 (0.2)	0	0
Hemorrhage intracranial	1 (0.2)	1 (0.2)	0	0

Hepatic cirrhosis	1 (0.2)	0	0	0
Hepatic failure	0	0	1 (0.2)	1 (0.2)
Hip fracture	1 (0.2)	1 (0.2)	0	0
Hypokalemia	1 (0.2)	1 (0.2)	0	0
Hydronephrosis	0	0	1 (0.2)	1 (0.2)
Hyperbilirubinemia	1 (0.2)	1 (0.2)	0	0
Influenza-like illness	1 (0.2)	0	0	0
Large intestinal ulcer perforation	1 (0.2)	1 (0.2)	0	0
Mental status changes	1 (0.2)	0	0	0
Musculoskeletal stiffness	0	0	1 (0.2)	0
Ischemic cardiac events§	1 (0.2)	1 (0.2)	0	0
Osteoarthritis	0	0	1 (0.2)	1 (0.2)
Osteoporosis	1 (0.2)	0	0	0
Pathological fracture	0	0	1 (0.2)	1 (0.2)
Peripheral edema	1 (0.2)	0	0	0
Peripheral swelling	0	0	1 (0.2)	1 (0.2)
Proctalgia	1 (0.2)	0	0	0
Renal injury	0	0	1 (0.2)	0

Rheumatoid arthritis	1 (0.2)	1 (0.2)	0	0
Seizure	1 (0.2)	0	0	0
Stomatitis	1 (0.2)	0	0	0
Subarachnoid hematoma	1 (0.2)	1 (0.2)	0	0
Subarachnoid hemorrhage	1 (0.2)	0	0	0
Subdural hemorrhage	0	0	1 (0.2)	1 (0.2)
Syncope	1 (0.2)	1 (0.2)	0	0
Thrombocytopenia	1 (0.2)	0	0	0
Tongue ulceration	1 (0.2)	0	0	0
Urinary incontinence	1 (0.2)	0	0	0
Urinary tract obstruction	1 (0.2)	1 (0.2)	0	0
Urethral stenosis	0	0	1 (0.2)	0
Vertigo	1 (0.2)	1 (0.2)	0	0
Weight increased	0	0	1 (0.2)	0

* Rash was a grouped term including eczema, eczema nummular, exfoliative rash, generalized rash, rash, dermatitis, dermatitis exfoliative, psoriasis, hyperhidrosis, maculo-papular rash, macular rash, drug eruption, erythema multiforme, lichenoid keratosis, pruritus, pruritic rash, skin exfoliation, skin lesion, skin reaction, and toxic skin eruption.

† New cancer neoplasms (benign, malignant, and unspecified [including cysts and polyps]) was a grouped term including malignant lung neoplasms, colon adenoma, colon neoplasms, leiomyosarcoma,

oropharyngeal cancer, papilloma, rectal adenocarcinoma, small-cell carcinoma, gastric

adenocarcinoma, colon adenocarcinoma, bladder cancer, and non-small-cell lung cancer.

‡ Infections was a grouped term including urinary tract infection, *Klebsiella* infection, sepsis, urosepsis, fungal infection, localized infection, lung infection, pneumonia, fungal respiratory tract infection, acarodermatitis, cellulitis, bacterial bronchitis, influenza, necrotizing fasciitis, and viral upper respiratory tract infection.

§ Ischemic cardiac events was a grouped term including myocardial infarction, acute myocardial infarction, acute coronary syndrome.

** Pain was a grouped term including pain in extremity, back pain, bone pain, upper abdominal pain, lower abdominal pain, musculoskeletal pain, neck pain, and noncardiac chest pain.

Supplementary Table S5. **Investigator-Reported Cause of Death on Study in the Safety Population.**

	Apalutamide	Placebo
	(n=524)	(n=527)
All deaths within 30 days of last dose – no. (%)	18 (3.4)	23 (4.4)
Death due to prostate cancer	8 (1.5)	7 (1.3)
Death due to adverse event	10 (1.9)	16 (3.0)
Adverse events leading to death – no. (%)		
Respiratory failure	1 (0.2)	2 (0.4)
Acute kidney injury	2 (0.4)	0
Acute myocardial infarction	1 (0.2)	1 (0.2)
Sudden death	0	2 (0.4)
Acute coronary syndrome	0	1 (0.2)
Cardiac failure	0	1 (0.2)
Cardio-respiratory arrest	1 (0.2)	0
Cardiogenic shock	1 (0.2)	0
Cerebrovascular accident	1 (0.2)	0
Death	0	1 (0.2)
Hypothermia	0	1 (0.2)
Intracranial hemorrhage	0	1 (0.2)

Large intestinal ulcer perforation	1 (0.2)	0
Myocardial infarction	1 (0.2)	0
Pulmonary embolism	0	1 (0.2)
Sepsis	0	1 (0.2)
Subdural hemorrhage	0	1 (0.2)
Sudden cardiac death	1 (0.2)	0
Suicide	0	1 (0.2)
Urosepsis	0	1 (0.2)
Vascular rupture	0	1 (0.2)

Supplementary Table S6. **Estimates of Subdistribution Hazard Ratio of the Apalutamide Group**

Relative to the Placebo Group. A post-hoc analysis accounting for the competing risk of death was

performed based on the Fine and Gray model.³ The estimates of a subdistribution hazard ratio of the

apalutamide group relative to the placebo group, along with the respective 95% confidence limits and the

P values from the Wald Chi-Square test, are presented. The results of this post-hoc analysis supported and

confirmed the preplanned analysis results that are currently presented in the manuscript.

			95% Confidence Limits of the	
	P Value	Hazard Ratio	Hazard Ratio	
Time to cytotoxic chemotherapy	<0.0001	0.408	0.286	0.581
Time to pain progression	0.1540	0.843	0.667	1.066
Time to chronic opioid use	0.2004	0.789	0.549	1.134
Time to skeletal-related event	0.2759	0.818	0.570	1.174

SUPPLEMENTARY REFERENCES

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